GATE

ALL BRANCHES

General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 13

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Recap of Previous Lecture

Topic Mensuration 2D & 3D

Topics to be Covered

Topic-1 Data Interpretation

[MCQ]

#Q. The diagonal of a rectangle is $\sqrt{41}$ cm and its area is 20 sq cm. The perimeter of the rectangle is?

$(a+b)^2 = a^2+b^2+2ab$

$(a+b)^2 - 2ab = a^2+b^2$

$P = 2(l+b)$

A 9 cm

B 18 cm

C 20 cm

D 41 cm

Area $= l \times b$

$20 = lb$

$(\sqrt{41})^2 = l^2 + b^2$

$41 = l^2 + b^2$

$\Rightarrow 41 = (l+b)^2 - 2 \times l \times b$

$\Rightarrow 41 = (l+b)^2 - 2 \times 20$

$\Rightarrow 81 = (l+b)^2$

$9 = l+b$

$P = 9 \times 2 = 18$ cm

Data Interpretation

Graph

Types:

The numerical data pertaining to any event can be presented by any one or more of the following methods.

1) Tables
2) Line Graphs
3) Bar Graphs or Bar Charts
4) Pie Charts or Circle Graphs

Line Graph:

Study the following graph carefully and answer the questions:

Marks scored by a student out of 500 in the Unit test held at different months of academic year 2019

[MCQ]

#Q. What is the percentage of marks scored by the student in the unit test of August and October taken together?

370 + 385

= 755

A 50%

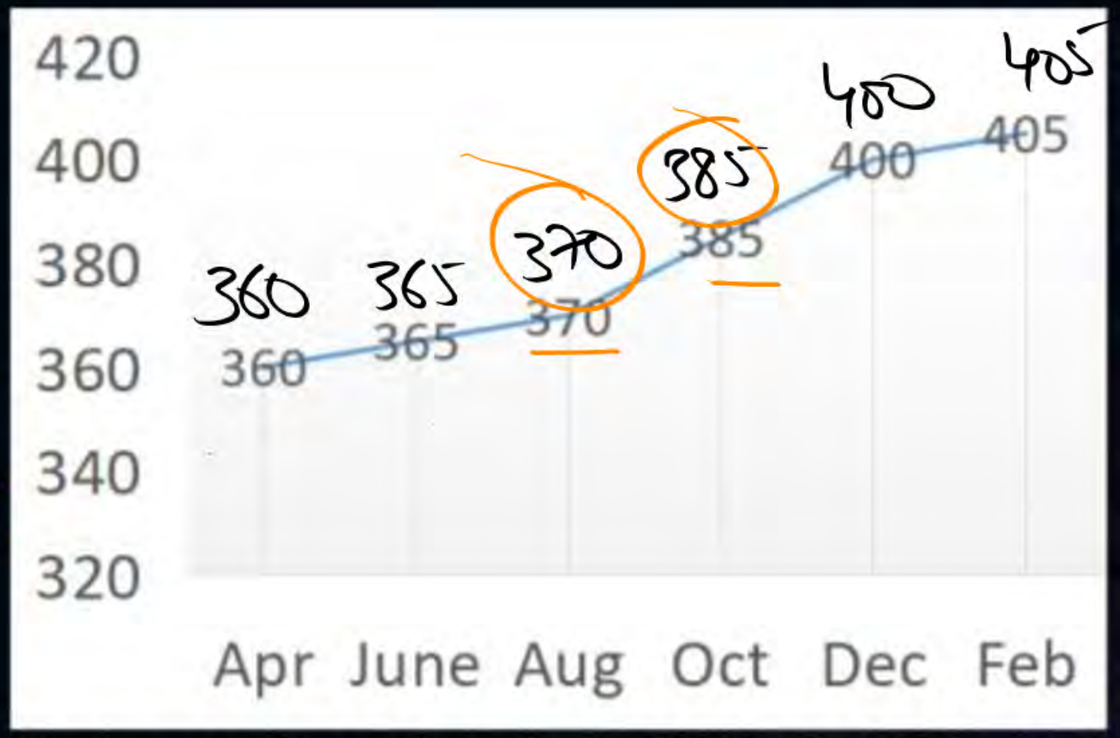
B 83.3%

C 60%

D 75.5%

$\% = \frac{755}{1000} \times 100$

= 75.5%



Marks scored by a student out of 500 in the Unit test held at different months of academic year 2019

[MCQ]

#Q. What are the average of marks obtained by the student in the unit test of given academic year?

$$\frac{725+755+805}{6}$$

A. 361.6

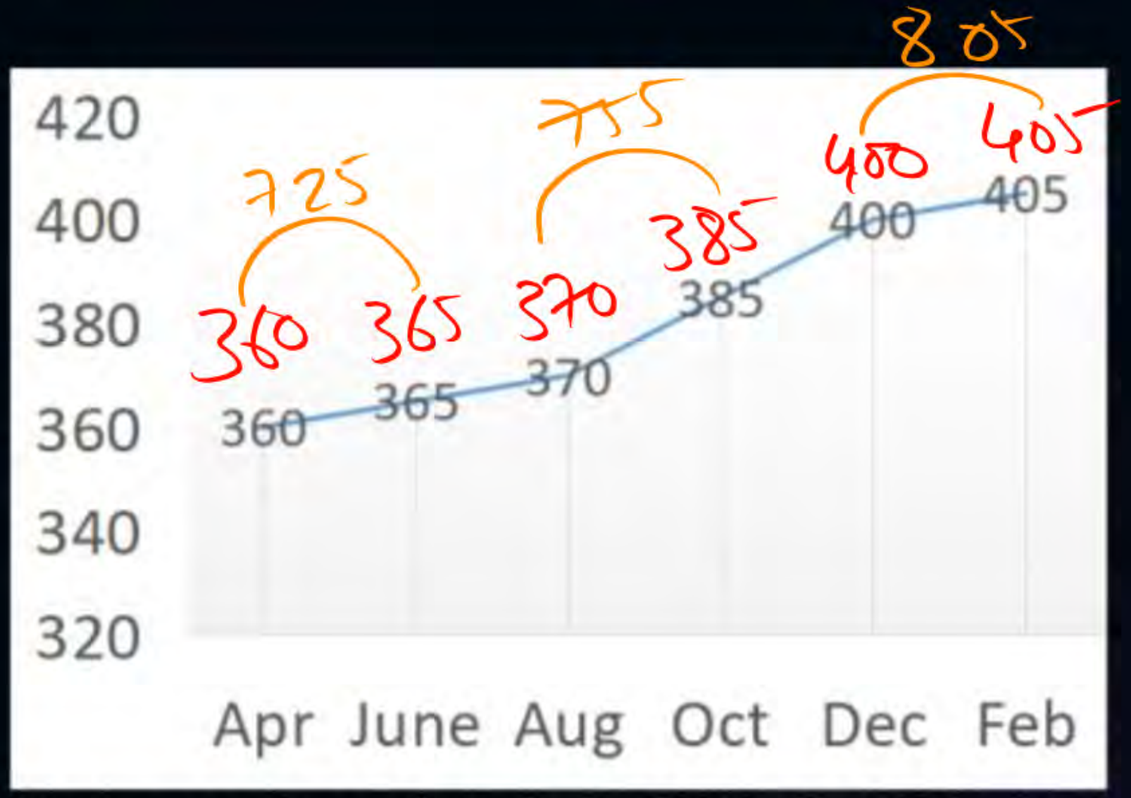
B. 380.8

C. 377.5

D. 400.0

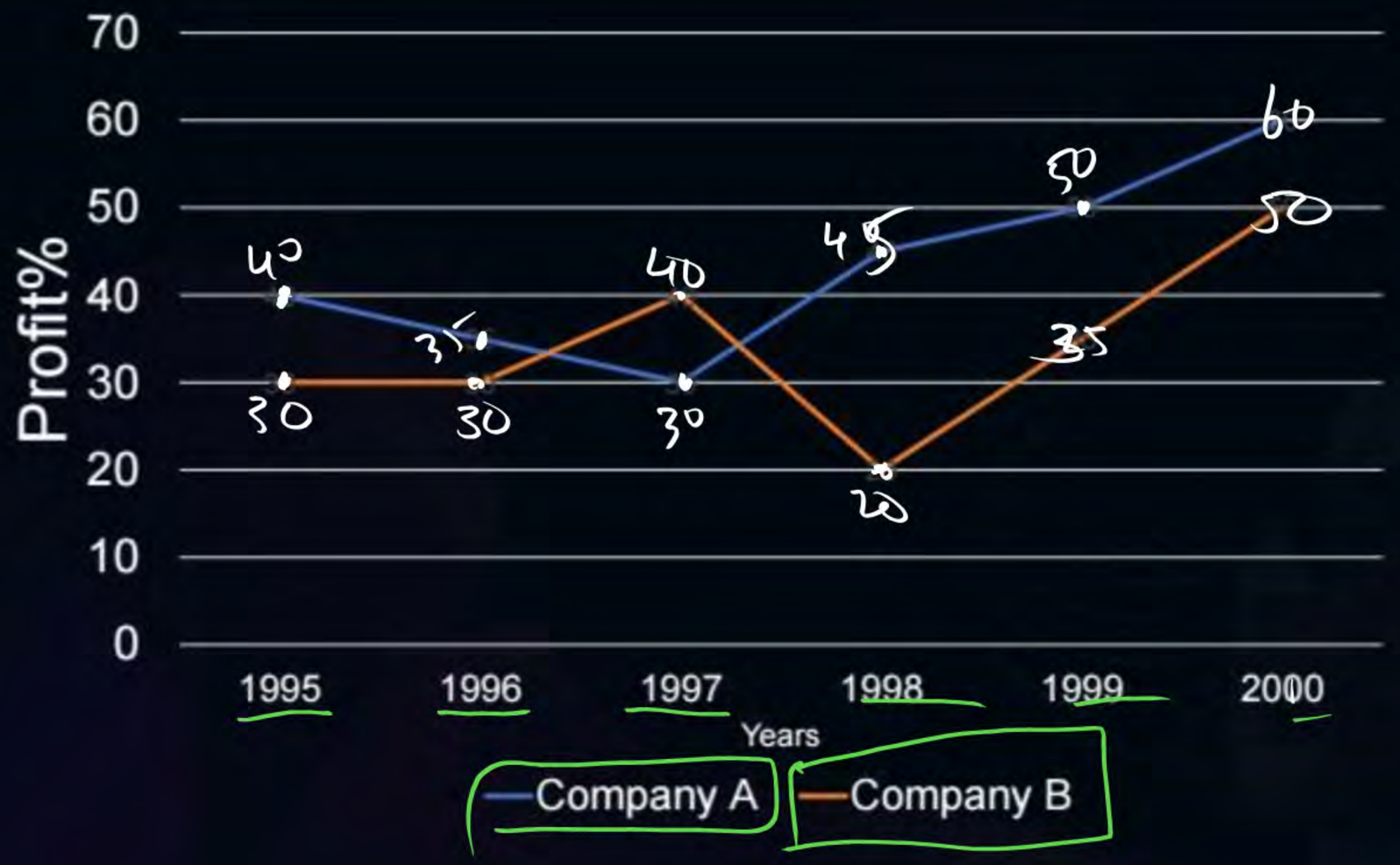
$$= \frac{2285}{6}$$

$$= 380.8\overline{3}$$



Marks scored by a student out of 500 in the Unit test held at different months of academic year 2019

Study the following table and answer the questions:



[MCQ]

#Q. If the expenditure of the company B in the year 1998 was ₹200 crore, what was its income?

$\frac{S.P}{C.P} = 1.2$

A. 360 crores

B. 300 crores

C. 240 crores

D. 120 crores

$\frac{S.P}{200} = 1.2$

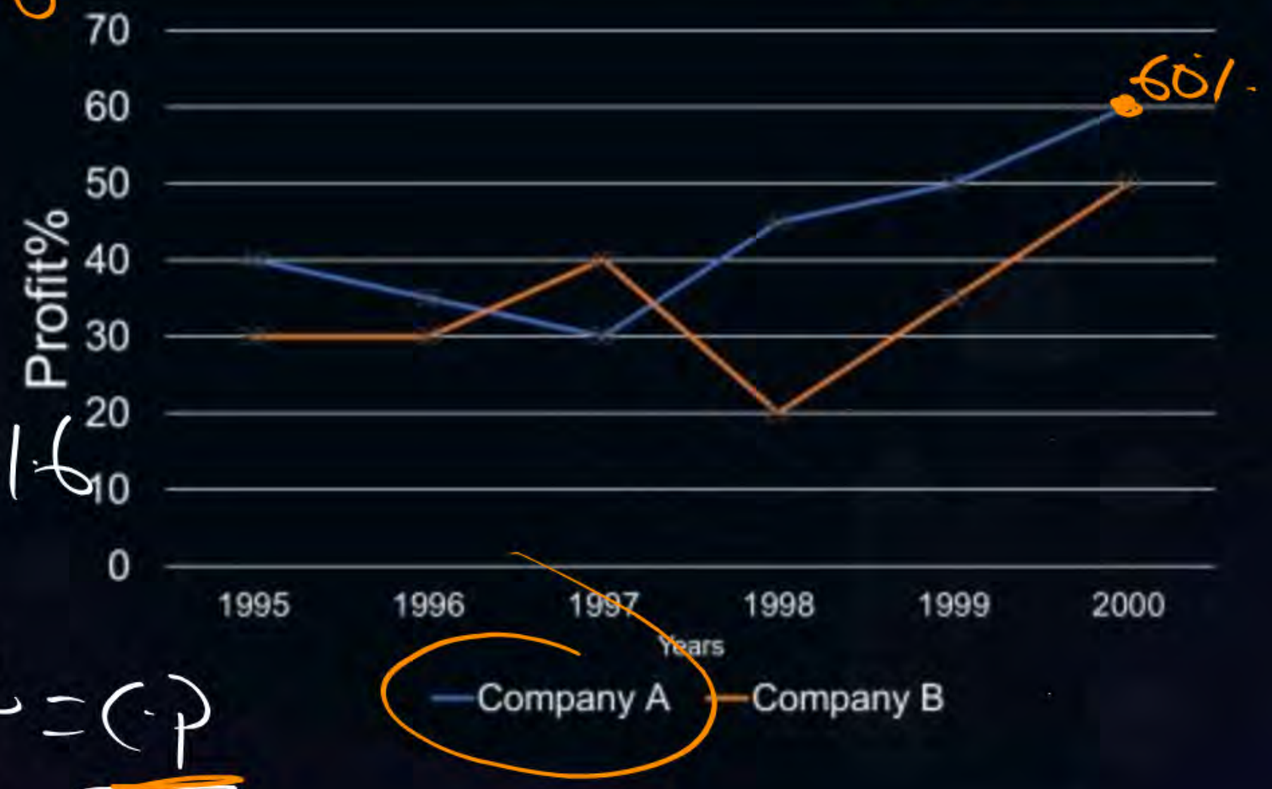
S.P = 240 crores

[MCQ]

#Q. If the income of company A in the year 2000 was ₹600 crore, what was its expenditure?

$\frac{S.P}{C.P} = 1.6$

375 crore



A. 210 crores

B. 275 crores

C. 340 crores

D. 375 crores

$\frac{600}{C.P} = 1.6$

$\frac{600}{1.6} = C.P$

[MCQ]

#Q. If the income of the two companies were equal in 1996, what was the ratio of their expenditure?

A ⇒ 35%. | B ⇒ 30%. $\frac{C.P_A}{C.P_B} = \frac{1.3}{1.35}$

(A) $\frac{S.P}{C.P} = 1.35$

⇒ $S.P = 1.35\,C.P_A$

(B) $S.P = 1.3\,C.P_B$

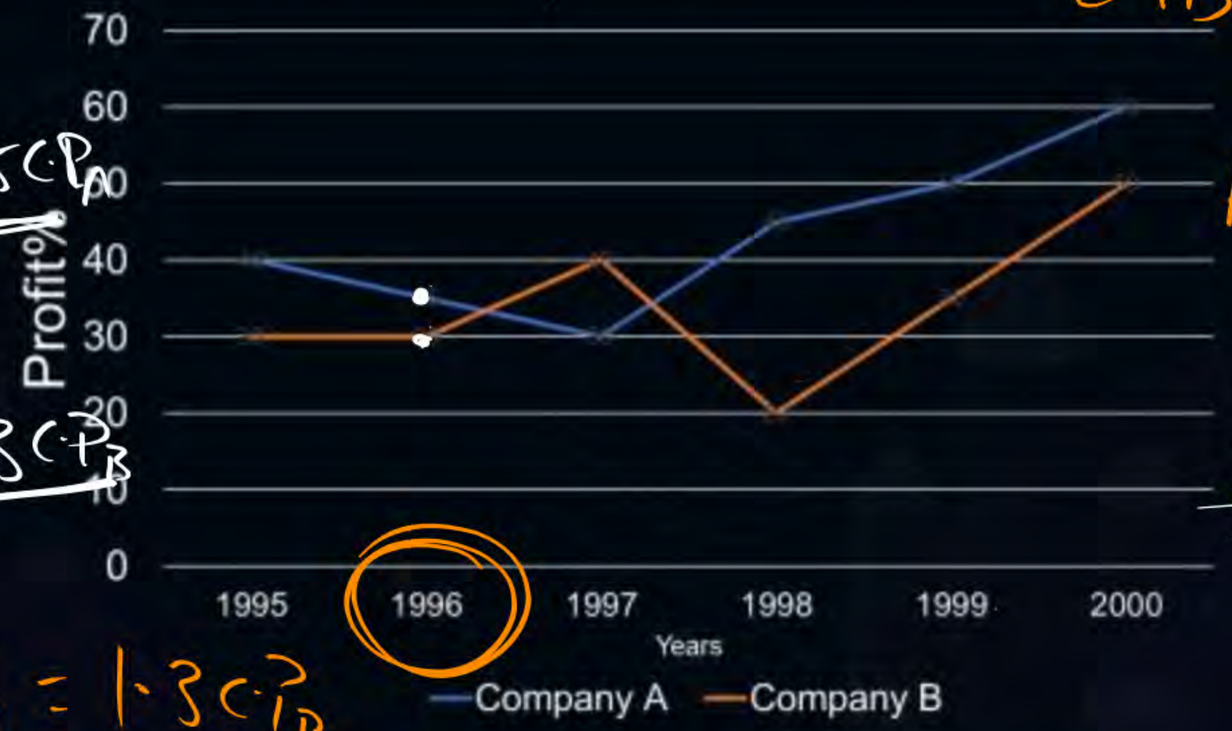
$1.35\,C.P_A = 1.3\,C.P_B$

A. 26 : 27

B. 100 : 67

C. 1 : 2

D. 5 : 27



$A : B = \cancel{130} : \cancel{135}$ (26, 27)

26 : 27

The following graph depicts the number of students who joined and left the school in the beginning of the academic year.

2005 → 3000

[MCQ]

#Q. By approximately what percent, the strength of school increased or decreased from 2007 to 2008?

2005 → 3000

3100 2950 3000 3150 3100 3200

A 1.2%

B 1.7%

C 2.1%

D 2.4%

$\frac{50}{2950} \times 100$

$= \frac{5000}{2950}\%$

$= 1.69\%$

$= 1.7\%$

[MCQ]

#Q. How many students were studying in the school during 2009?

3150

A 2950

B 3000

C 3100

D 3150

[MCQ]

#Q. The number of students studying in school in 2008 was what percent of the number of students studying in the scool in 2011?

$\frac{3000}{3200} \times 100$

1500
16

3000 3200

A 92.13%

B 93.75%

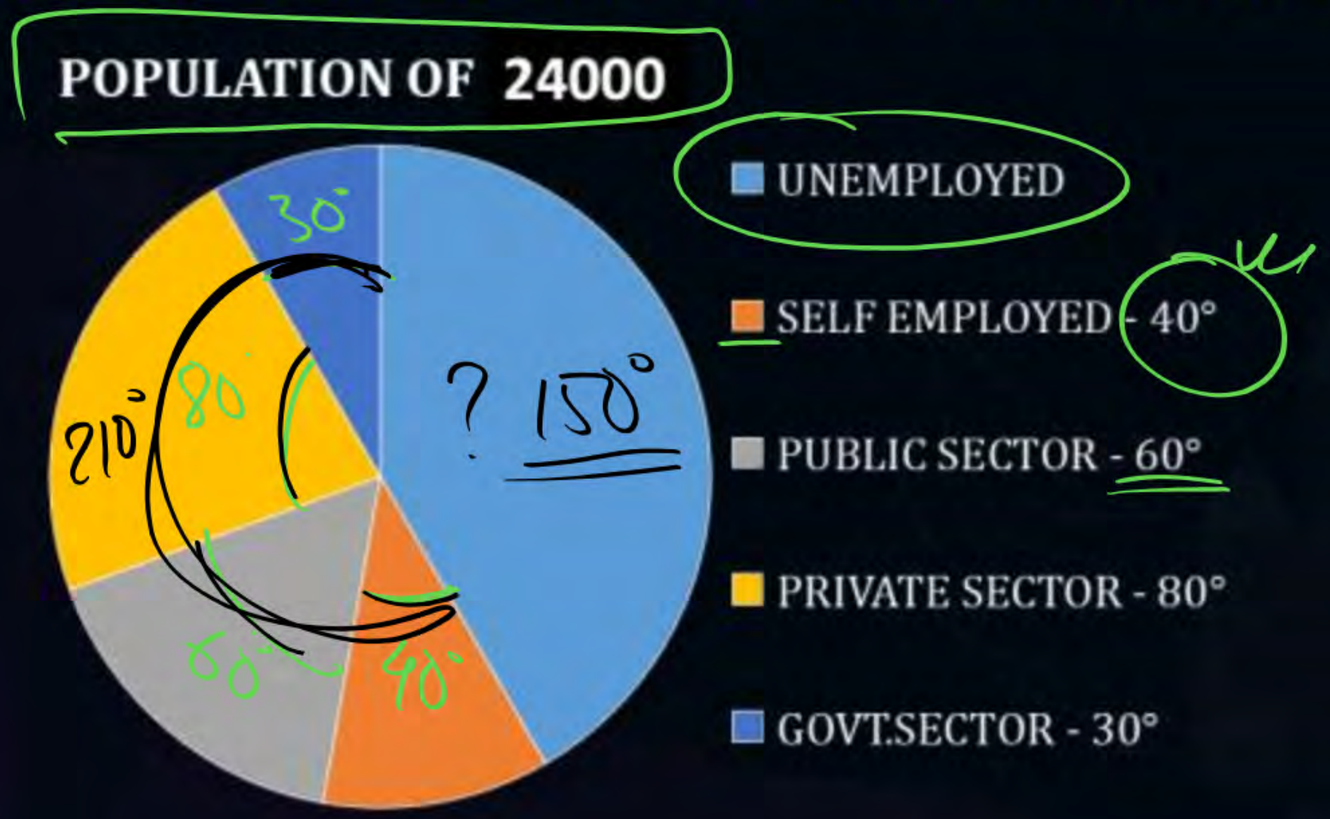
= 93.75%

C 97.25%

D 96.88%

Pie Chart:

Study the following graph carefully and answer the questions:

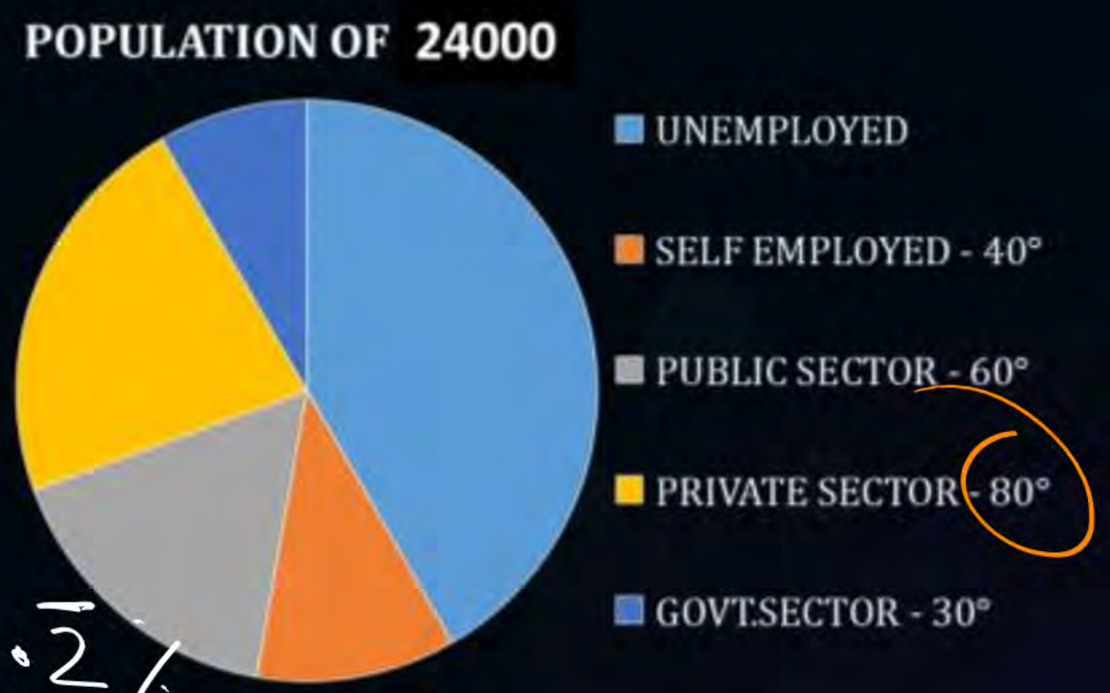


[MCQ]

#Q. If private sector employees are to be mentioned in ~~above~~ below chart in the form of percentage, it would be .....%.

$$\frac{80^\circ}{360^\circ} \times 100$$

$$= \frac{200}{9}\% = 22.\overline{2}\%$$

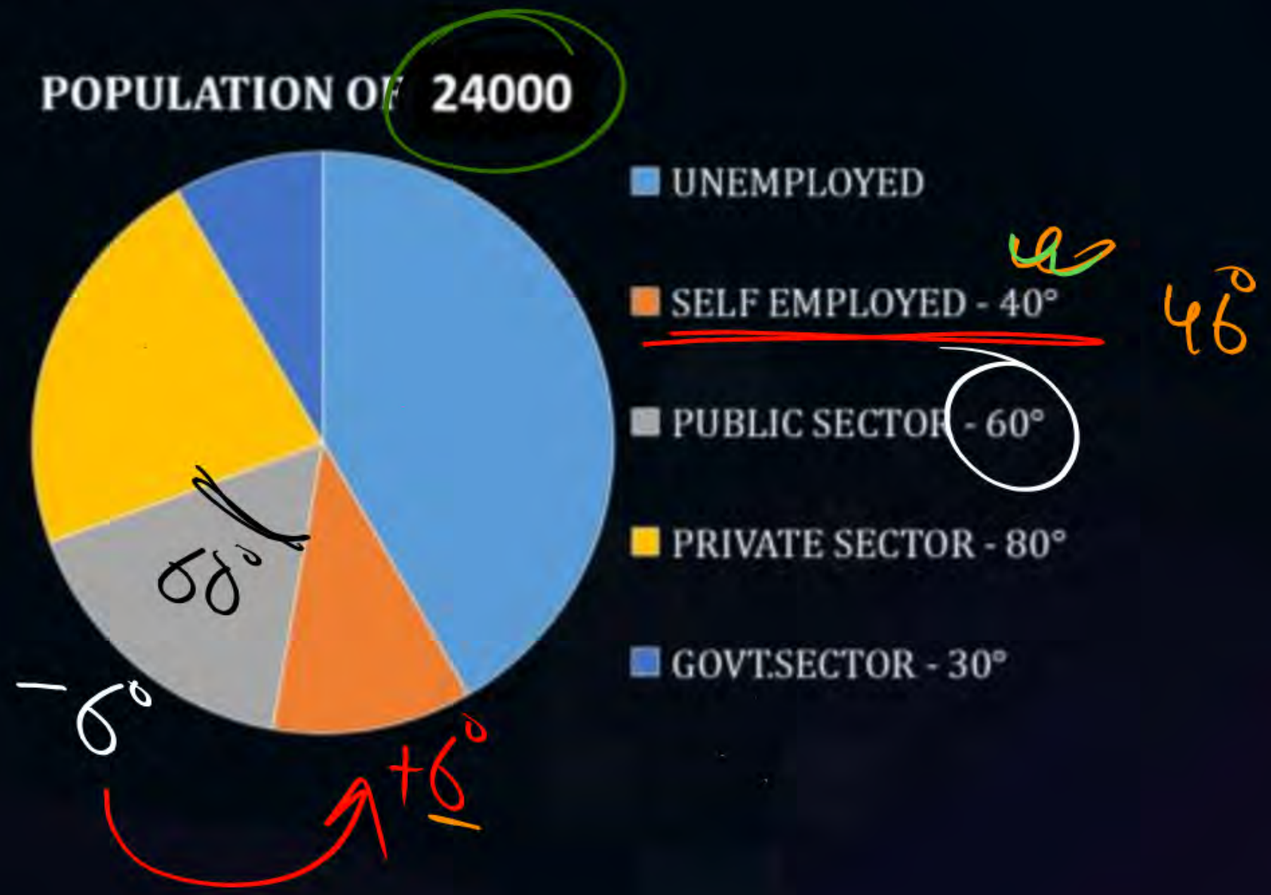


[MCQ]

#Q. If 10% of public sector people resign and becomes self-employed, find the increase percent in self-employed population.

$\frac{46}{40} = 1.15$

$\frac{6}{40} \times 100 = 15\%$

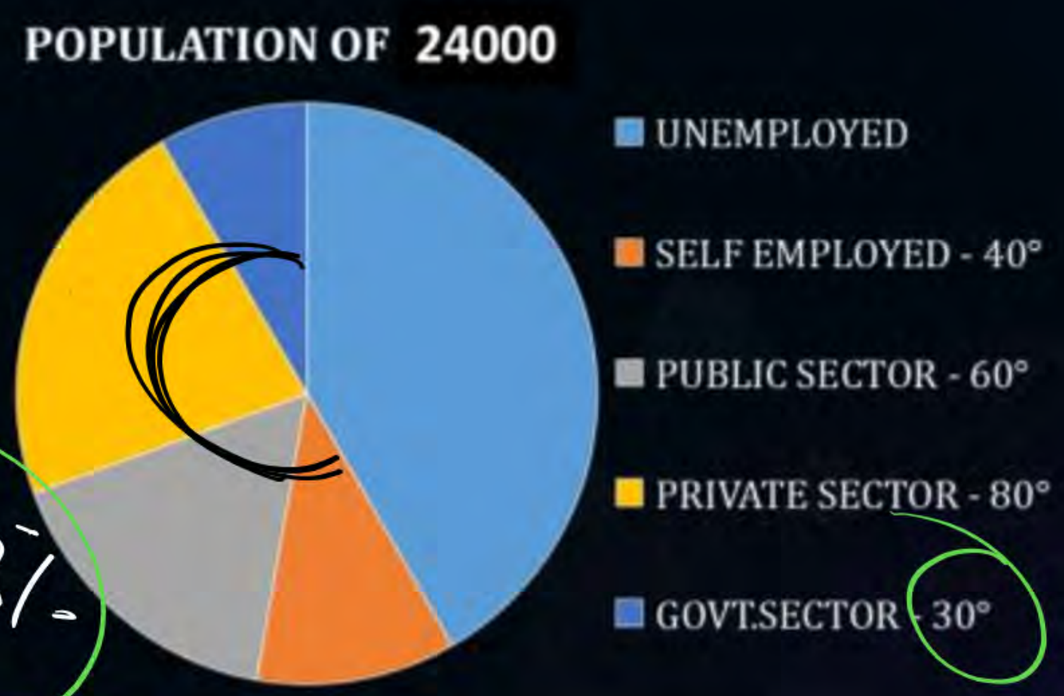


[MCQ]

#Q. What percentage of employed people are working in government sector?

$$\frac{30^\circ}{210^\circ} \times 100$$

$$= \frac{100}{7} = 14.28\%$$



[MCQ]

#Q. Find the number of people who are unemployed.

$\frac{150}{360} \times 24000$

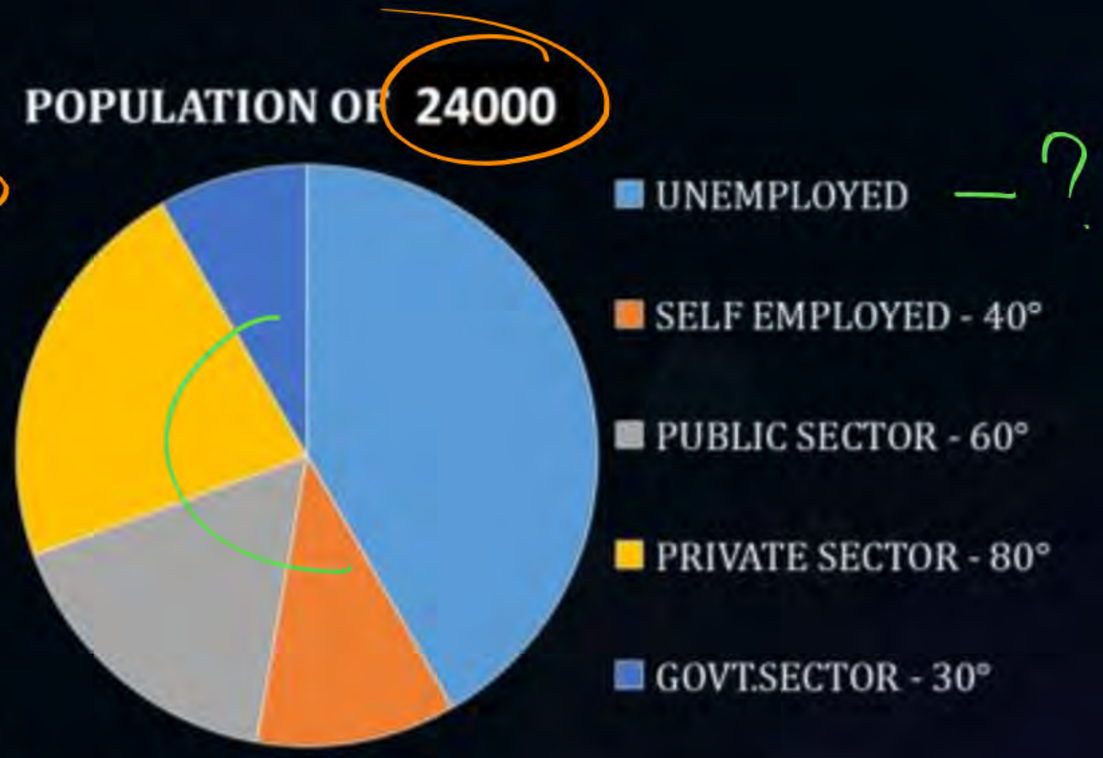
5

2000

12

1

= 10,000



Tabular form:

Study the following table and answer the questions:

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

[MCQ]

#Q. Who among the following employees earns maximum bonus in comparison to his total income?

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

A. P

B. Q

C. R

D. S

$\frac{80}{900} \times 100$

$P = \frac{80}{9} = 8.\overline{8}\%$

$Q = \frac{40}{5}\% = 8\%$

$R = 10\%$

$S = \frac{80}{7} = 11.42\%$

$T = 10\%$

[MCQ]

#Q. The income from overtime is what percent of income from arrears in the case of employee P?

A. 90

B. 80

C. 40

D. 75

$\frac{180}{200} \times 100$

90

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

[MCQ]

#Q. How many employee have their salary more than 3 times the income from bonus?

Assignment

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

A. One

B. Five

C. Three

D. None

[MCQ]

#Q. Who among the following employees has maximum percentage of his salary out of the total income?

Assignment

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

A. T

B. P

C. S

D. R

[MCQ]

#Q. Who among the following employees has minimum ratio of income from arrears to the income from salary?

Assignment

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

A. P

B. Q

C. T

D. S

2 mins Summary

Topic Data Interpretation

THANK - YOU